

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM watsonx.ai (Granite)”

Project Tile – Interview Trainer AI Agent

Domain of Project – Education

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Problem Statement -

Problem Statement : Interview Trainer AI Agent

The Challenges

- **Generic AI Responses:** LLMs may generate inaccurate or hallucinated interview answers.
- **Lack of Personalization:** Generic interview questions ignore company-specific requirements.
- **Difficult Answer Evaluation:** Scoring technical and behavioral responses objectively is challenging.
- **Learning Gap Identification:** Tracking weak areas and generating personalized study plans is complex.

The Objectives

Build an **AI-powered Interview Trainer Agent** that delivers **personalized interview preparation, AI-driven evaluation, and career guidance for students and professionals**. The solution should:

- **Personalized Interview Preparation** – Generate role-specific and company-specific interview questions based on job role, experience level, and interview category (Technical, HR, Behavioral, System Design).
- **RAG-based Knowledge Retrieval** – Retrieve relevant interview concepts, coding patterns, and company-specific information using ChromaDB and Retrieval-Augmented Generation (RAG) for context-aware responses.
- **AI Answer Evaluation & Feedback** – Evaluate candidate answers on accuracy, clarity, confidence, and completeness while providing instant feedback and improvement suggestions.
- **Personalized Learning Roadmap** – Identify weak areas from interview performance and generate customized learning plans to improve technical and communication skills.
- **Multi-Agent System**
 - **Question Generation Agent** – Generates personalized interview questions.
 - **Knowledge Retrieval Agent** – Retrieves relevant interview knowledge using RAG.
 - **Answer Evaluation Agent** – Scores responses and provides AI-powered feedback.
 - **Learning Roadmap Agent** – Creates personalized study plans based on performance.
 - **Performance Analytics Dashboard** – Visualize interview scores, category-wise strengths & weaknesses, progress trends, and overall interview readiness through interactive charts.

Proposed Solution -

Proposed Solution: AI-Powered Interview Trainer Agent

- **Context-Aware Question Generation:** Uses IBM watsonx.ai and Retrieval-Augmented Generation (RAG) with ChromaDB to generate accurate, role-specific, and company-specific interview questions.
- **AI-Powered Answer Evaluation:** Evaluates candidate responses on **Accuracy, Clarity, Completeness, and Confidence**, providing instant feedback and improvement suggestions.
- **Company-Specific Interview Preparation:** Delivers customized interview experiences for companies like Google, Amazon, Microsoft, Infosys, and TCS using dedicated knowledge bases.
- **Personalized Learning Roadmaps:** Identifies weak areas from interview performance and automatically generates personalized study plans.
- **Secure & Scalable Architecture:** Built using **React, FastAPI, MongoDB Atlas, ChromaDB** (Persistent Vector Store), **IBM watsonx.ai (Granite)**, **Render**, and **Vercel** for a scalable cloud-native deployment.to provide a responsive, secure, and scalable interview preparation platform.
- **Interactive Performance Dashboard:** Visualizes interview scores, strengths, weaknesses, progress trends, and readiness through analytics dashboards.

Technology Used

1. AI & LLM

IBM watsonx.ai (Granite Foundation Model): Interview question generation, answer evaluation, personalized feedback, and roadmap generation.

ChromaDB: Vector database for semantic search and Retrieval-Augmented Generation (RAG).

2. Backend

FastAPI (Python): High-performance REST API framework.

MongoDB Atlas: User authentication, interview history, and analytics.

JWT Authentication: Secure user authentication.

Bcrypt (Passlib): Password hashing and security.

3. Frontend

React.js + Vite: Fast and modern UI development.

React Router DOM: Client-side routing.

Axios: Frontend–backend API communication.

Tailwind CSS: Responsive UI design.

Recharts: Performance analytics and visualization.

4. Cloud & Deployment

IBM watsonx.ai – AI model inference using Granite Foundation Models.

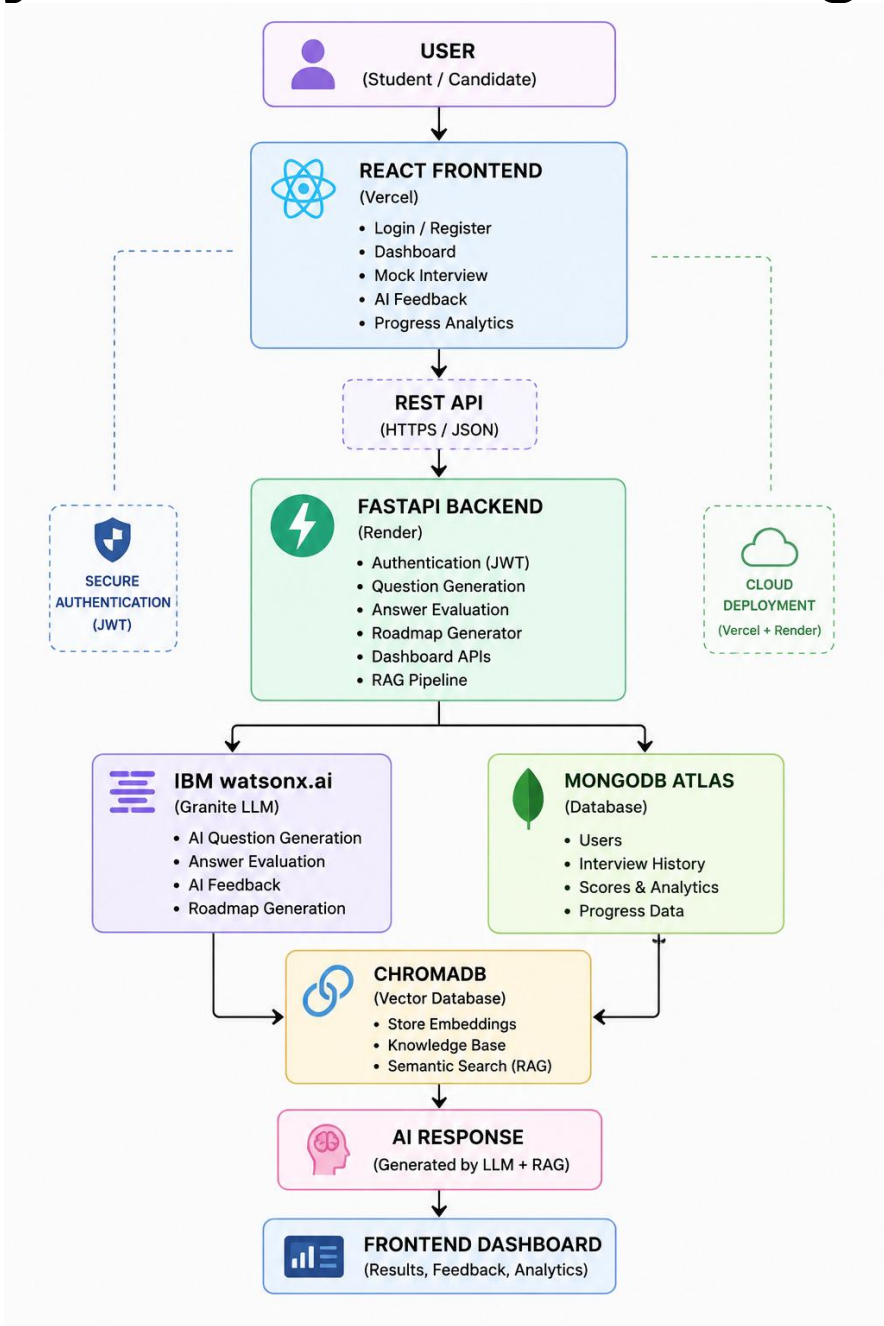
IBM Cloud Object Storage (COS) – Storage for project resources.

Docker – Containerized backend deployment.

Render – Backend deployment (FastAPI).

Vercel – Frontend deployment (React + Vite).

System Architecture Diagram



Role of Agentic AI in the solution

Role of Agentic AI in Interview Trainer AI Agent

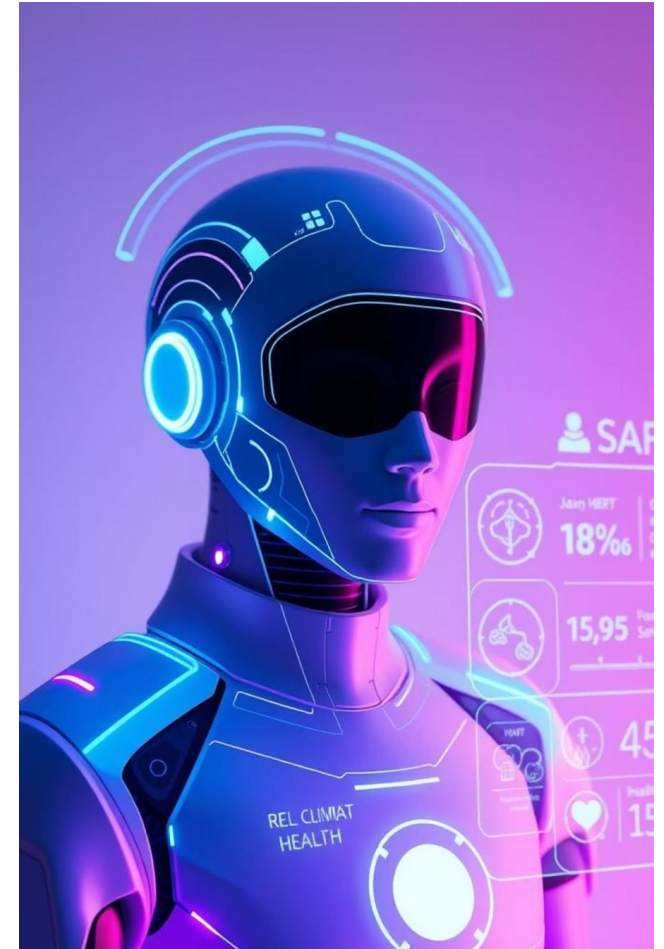
Agentic AI enables the **Interview Trainer AI Agent** to function as an intelligent, adaptive interview assistant rather than a simple question-answering chatbot. Using **IBM watsonx.ai (Granite Foundation Model)** and a **Retrieval-Augmented Generation (RAG)** pipeline, it delivers personalized interview preparation and real-time guidance.

The system coordinates multiple specialized AI agents, each responsible for a specific task such as generating interview questions, retrieving relevant interview knowledge, evaluating candidate responses, and creating personalized learning roadmaps. These agents work collaboratively to provide accurate, context-aware, and actionable feedback.

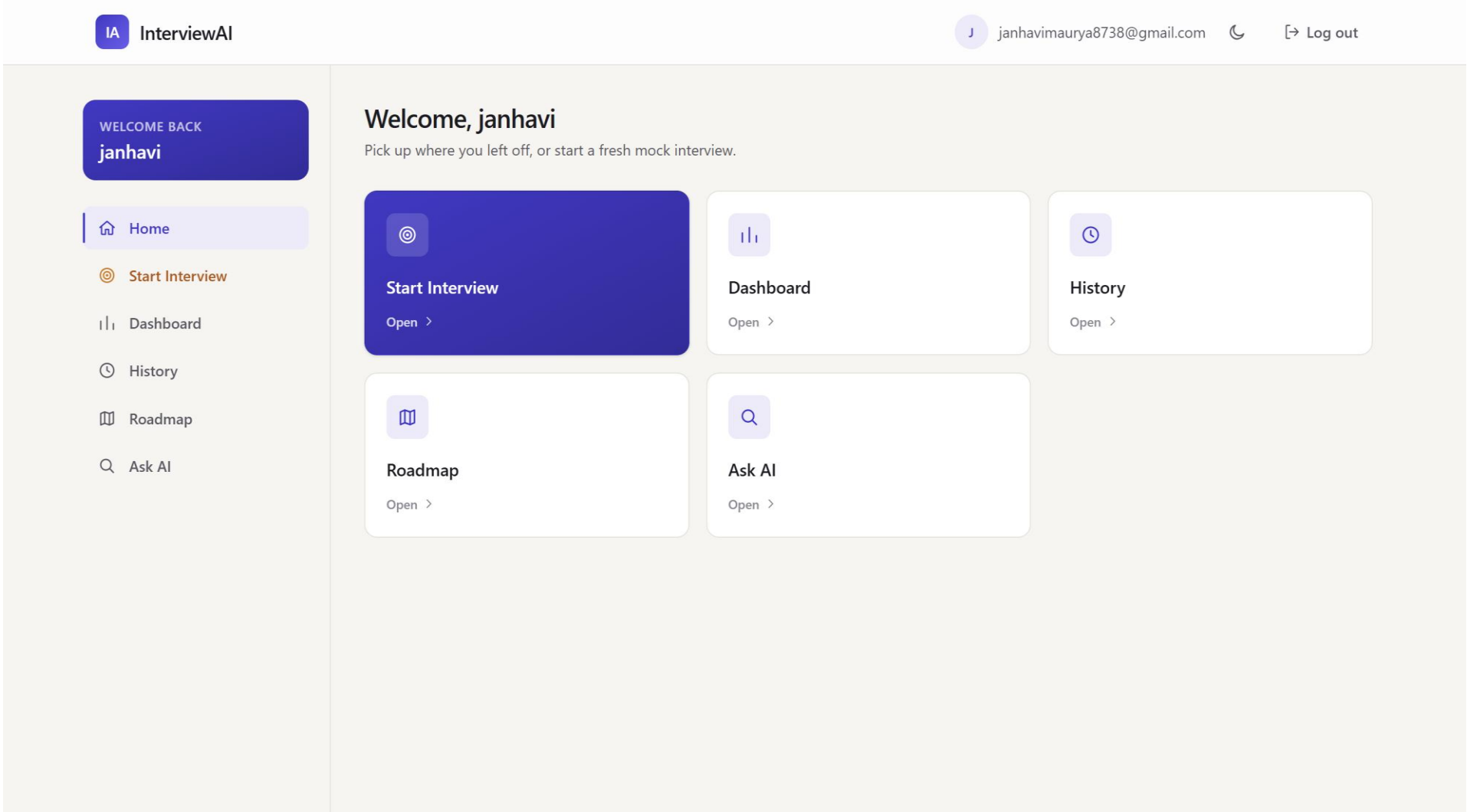
Agentic AI understands the user's profile, including **job role, experience level, target company, and interview category**, to generate customized interview questions and recommendations. It continuously adapts to user performance by identifying strengths and weak areas, ensuring a personalized learning experience.

Additionally, the agents share contextual information and evaluation results to improve decision-making, enabling more relevant questions, better answer evaluation, and dynamic study plans.

Overall, Agentic AI makes the Interview Trainer **intelligent, proactive, personalized, and goal-driven**, acting as a smart AI interview coach that helps users improve their technical, behavioral, and problem-solving skills while increasing their interview readiness.



Some Screenshots-



Interview Setup

Configure your practice session or choose a company-specific prep mode.

Choose a Company (Optional)



General Prep

Mixed questions across all domains



Google

Focus on distributed systems, scale, and behavioral fit.

System Design

Scalability



Amazon

Heavy behavioral focus on 16 Leadership Principles.

Leadership Principles

System Design



Microsoft

Values growth mindset, collaboration, and Azure knowledge.

Growth Mindset

System Design



TCS

Focus on basics, stability, and process adherence.

Fundamentals

Aptitude



Infosys

Values ethics, learning agility, and quality focus.

CLIFE Values

Fundamentals

Session Configuration

Target Role

Software Engineer

Experience Level

Mid (3-5 yrs)

Questions

5

3

15

Start Interview Session

Amazon Interview

1 of 5

system_design

Hard

Rate Limiting

Design an API rate limiter that allows X requests per Y time window from each user while handling distributed systems challenges like consistency and scalability.

Hint: Consider data structures like token buckets or leaky buckets along with Redis or DynamoDB for storage.

Your Answer

Fault Tolerance: Use Redis replication and clustering for high availability.

Example

Limit: 100 requests per minute per user

User sends 101st request within the same minute → HTTP 429 Too Many Requests

Counter resets after 60 seconds.

Tech Stack: API Gateway + Redis + Load Balancer + Application Servers.

This design provides low latency, scalability, and consistency in distributed environments.

AI is evaluating...

Tech Stack: API Gateway + Redis + Load Balancer + Application Servers.

This design provides low latency, scalability, and consistency in distributed environments.

Next Question →

SCORE

84/100

Accuracy

88

Clarity

84

Completeness

80

Confidence

86

Feedback

Good understanding of core concepts but could elaborate on fault tolerance mechanisms.

✧ Model Answer

Use a token bucket algorithm stored in a distributed cache such as Redis Cluster. Each user has a unique identifier; incoming requests decrement tokens atomically using Lua scripting to prevent race conditions across nodes. Configure Redis with synchronous replication and automatic failover for high availability. Scale horizontally by adding more shards and balancing traffic through a consistent hashing layer at the API gateway. This ensures sub-millisecond latency under peak loads while maintaining strict per-user limits.

Performance Dashboard

Total Sessions

1

Questions Answered

11

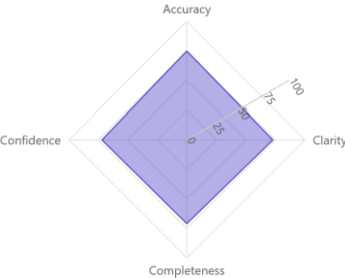
Average Score

73

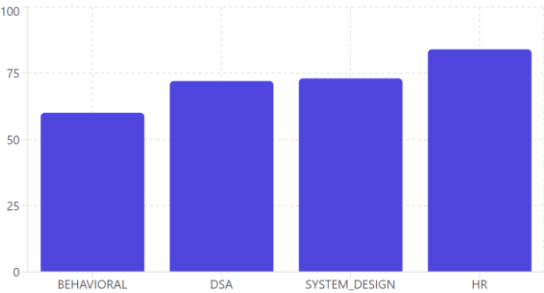
Best Score

87

Skill Breakdown



Category Performance



Score Trend



Personalized Study Roadmap

7-day roadmap generated!

Generate a personalized study plan based on your weak areas and past performance.

Duration: 7 Days 15 Days

Generate 7-Day Plan

Your 7-Day Study Plan

Focus: dsa, behavioral
Estimated Time: 14 hours

DAY 1

DSA Fundamentals

DSA

Tasks

- Learn Arrays & Strings
- Solve 5 Easy LeetCode Problems
- Revise Time Complexity
- Write notes

Resources

~2h

AI Interview Assistant

Ask interview questions and get AI-powered answers instantly.

Your Question

Explain Binary search

Categories

- All Categories
- DSA
- HR
- Behavioral
- System Design

Ask AI

AI Answer

Copy

- Describe what binary search is conceptually
- Outline its algorithmic approach
- Discuss time/space complexities
- Provide pseudocode/actual code example
- Mention variations/applications
- Highlight when it can/cannot be applied

Binary search is a fundamental algorithm widely used for efficiently locating elements within sorted collections. Its conceptual foundation lies in repeatedly dividing the search interval in half, thereby narrowing down the possible locations where the target value may reside. By leveraging the ordered nature of the data, binary search achieves a time complexity of $O(\log n)$, making it significantly faster than linear search for large datasets.

Novelty and Uniqueness

- **AI-Powered Personalized Interview Preparation**

Generates personalized interview questions based on the user's **role, experience level, target company, and interview category**.

- **Retrieval-Augmented Generation (RAG)**

Uses **ChromaDB + IBM watsonx.ai (Granite Foundation Model)** to retrieve verified interview knowledge, reduce hallucinations, and generate context-aware responses.

- **Intelligent AI Performance Evaluation**

Evaluates candidate responses across **Accuracy, Clarity, Completeness, and Confidence**, providing instant AI-powered feedback and improvement suggestions.

- **Company-Specific Interview Preparation**

Delivers customized interview experiences for companies such as **Google, Amazon, Microsoft, Infosys, and TCS** using dedicated knowledge bases.

- **AI-Driven Personalized Learning Roadmap**

Identifies weak areas from interview performance and generates customized learning paths with topic recommendations and practice plans.

- **Cloud-Native AI Deployment**

The complete solution is deployed using **Vercel** (Frontend) and **Render** (Backend), enabling real-time access without local setup.

Git Hub Link

GitHub Repository & Live Demo

- **GitHub Repository**

Repository Name: Interview-Trainer-AI

GitHub Link: <https://github.com/janhavi-2011/Interview-Trainer-AI.git>

- **Live Demo:** <https://interview-trainer-ai-three.vercel.app>

Backend API : <https://interview-trainer-ai.onrender.com/docs>

Future Scope

- **Voice-Based Mock Interviews**

Integrate speech recognition and text-to-speech to enable real-time AI interview conversations.

- **AI Interview Avatar**

Introduce a virtual AI interviewer with facial expressions, natural interaction, and real-time feedback.

- **Multi-Language Support**

Support interview preparation in multiple languages such as English, Hindi, and other regional languages.

- **Advanced Performance Analytics**

Provide detailed skill tracking, progress reports, and personalized improvement recommendations.

- **Cloud-Native Enterprise Deployment**

Migrate the platform to IBM Cloud Kubernetes or Code Engine with auto-scaling, load balancing, and enterprise-grade monitoring for large-scale production deployments.

- **Integration with Job Portals**

Connect with LinkedIn, Naukri, and other job portals to provide role-specific interview preparation and career recommendations.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Janhavi Maurya

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Learning hours: 30 mins

Thank You!

Thank you for your time and interest.